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COS700 Research Proposal

Real-Time Behavioural Risk Scoring and Fraud Detection for Cardless Cash Withdrawal Fraud

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Abstract

TODO: After full implementation

Keywords:

TODO: After full implementation

1 Introduction

This research proposal details the investigation of how real-time behavioural risk scoring can be used to detect suspicious or fraudulent cardless cash withdrawal transactions in South African banking applications. The specific fraud scenario discussed in this research is social-engineering driven cardless cash withdrawal fraud. In this type of scam, a bad actor contacts a victim by phone and pretends to be their a banking institution, mobile service provider or some other trusted organisation to the victim. The victim is then pressured into opening their banking application and performing an action that the fraudster may claim is protective, but actually allows them to scam the user out of their money. The primary scam that will be considered involves the victim creating a cash-send or similar cardless withdrawal voucher and then sharing the voucher number, PIN or access code with the fraudster.

The main reason this fraud type will be investigated is that the transaction may look legitimate from the bank's point of view. The customer may be using their own phone, their own banking profile and their own authentication method. It is not a traditional manner of service compromise where an attacker has directly hacked into the banking application. In these types of cases, the customer is manipulated into carrying out the transaction themselves. This makes detection of the fraud difficult because OTPs maybe entered and transactions approved making the process seem normal. Therefore in order to address this type of scam the banking security systems have to consider and analyse the behaviour during the transaction session in addition to whether the transaction was correctly authorised.

This issue is highly relevant in the South African banking context. SABRIC's 2024 Annual Crime Statistics report detailed that digital banking fraud incidents increased from 52,584 in 2023 to 97,975 in 2024 (an increase of 86%), while gross losses increased from R1.082 billion to R1.888 billion, which was an increase of 74% (SABRIC, 2025). SABRIC also reported that banking applications were the dominant digital fraud channel in 2024, accounting for 65.3% of all reported incidents. Banking app fraud incidents also increased from 31,612 in 2023 to roughly 64,000 in 2024, with the report stating that these banking app incidents involved social engineering and human error rather than breaches of banking application platforms themselves (SABRIC, 2025). These results tie in directly to the proposed topic and highlights why fraud detection must expand to encompass non technical surfaces of attack.

Cardless withdrawal services are legitimate banking features that allow a customer to send money to another person, often using only a cellphone number, where the recipient can withdraw the money from an ATM without a physical bank card. Standard Bank Instant Money, Absa CashSend, Nedbank cardless withdrawal and FNB eWallet are examples of this type of service in South Africa. Kinsman (2019) explains cardless banking in the South African context and describes services such as Absa Cash Send and FNB eWallet as convenient and accessible services that can support financial inclusion for those that have not traditionally had access to modern banking technologies and helps bridge the gap of financial service access. Nambiar and Bolar (2023) also show that cardless cash technology is a meaningful banking service as many users prefer it

for its accessibility and simplicity over a traditional card-based cash withdrawal. The service itself is therefore very useful and and convenient, especially in countries that still primarily rely on cash payments over bank cards. However, this convenience also makes it a popular method of theft and fraud amongst criminals.

The proposed research assignment will therefore focus on designing and building a prototype that simulates real-time mobile banking events, extracts behavioural features, calculates a risk score and triggers an alert or warning when a cardless withdrawal event appears suspicious. It will investigate whether a combination of rule-based systems, anomaly detection and a hybrid approach can identify high-risk behavioural patterns that may justify temporary transaction ban pending additional user verification or even escalation to the bank's dedicated fraud team.

2 Problem Statement

Social-engineering scams such as "vishing" (voice phishing) can manipulate individuals into performing transactions that appear legitimate, but are in fact attempting to steal the user's money. In the case of cardless cash withdrawal fraud, the victim may be instructed to generate a voucher or cash-send transaction under the false belief that they are protecting their account, reversing fraud, verifying themselves or moving money to safety. Because the customer may be authenticated and may initiate the transaction themselves, traditional authentication controls may not be enough to identify that the transaction is being performed under duress.

A transaction can be technically valid, but still be fraudulent in nature. This is especially important for cardless cash withdrawal products because cash can be withdrawn quickly once the voucher or PIN have been shared. Once the money has been withdrawn, recovery almost impossible. Therefore the key detection time before or during cash redemption, rather than after the victim reports the scam.

The research question is therefore:

Can real-time behavioural risk scoring identify high-risk cardless cash withdrawal transactions associated with social-engineering scams in mobile banking?

The following sub-questions were also identified:

- Which behavioural indicators are relevant for detecting suspicious cardless cash withdrawal activity?
- How effective are rule-based, anomaly-based and hybrid risk scoring approaches at identifying high-risk cardless withdrawal transactions?
- What type of intervention or alerting system should be implemented when fraudulent activity is detected

3 Literature Study

3.1 Digital Banking Fraud and Social Engineering

Online banking has revolutionised the way people make and receive payments in the modern age. Services that were previously performed in a branch or via an ATM can now be performed through mobile applications and online banking platforms. This increases banking convenience, but it also increases the attack surface for digital fraud. Phiri, Lavhengwa and Segooa (2024) detail that online banking gives customers easy access to banking services, but that online banking fraud is common globally and causes damage to both banks and consumers.

As discussed in the previous section, the 2024 SABRIC figures with regards to digital fraud is particularly relevant to this research topic due to the fact that social engineering was the primary driver behind reported banking app fraud. SABRIC (2025) reported that criminals exploited human error through social engineering rather than attacking and breaching banking application platform used by consumers (SABRIC, 2025). The European Payments Council (2025) also reported that social engineering and phishing on rising threats and that social engineering increasingly leads to authorised push payment fraud, where the target is the user rather than the technology itself. Authorised payment fraud can therefore be defined as a situation where bank customers are tricked into transferring money from their own accounts under false pretences by bad actors (Braithwaite, 2024).

Salahdine and Kaabouch (2019) explain that that social engineering is an attack mechanism that bypasses the technical guards and controls enforced by banking institutions by going directly to the consumer. This affords them the opportunity to reduce the risk of being detected and get caught, lowers the technical expertise required and perform the scam multiple times with many different users. It is therfore crucial that fraud detection systems can no longer simply rely the customer successfully authenticated themselves with the system. A customer can authenticate successfully and still be acting under manipulation. This main reason why behavioural risk scoring can be useful. The system should look for behavioural signs that the transaction context is abnormal, rushed or inconsistent with the customer's normal pattern.

3.2 Cardless Cash Withdrawal Services

Cardless cash withdrawal services allow money to be made available for withdrawal without the recipient needing the sender's physical bank card. The exact implementation differs between banks, but the broad mechanism is similar. A sender creates a transaction through a mobile app, internet banking or other banking service. A voucher or withdrawal number is then generated along with a PIN or access code. This is then used by the recipient to withdraw the funds at an ATM or cash point as cash.

Kinsman (2019) explains that cardless banking as a financial inclusion tool for those that have previously not had access to modern banking features. Kinsman also goes onto say the ease and convenience of the technology is what makes it useful for "underbanked" consumers.

Various bank product pages further confirm how these services work in practice. Absa describes CashSend as a service that allows money to be sent to anyone and withdrawn immediately without the recipient needing a bank account or card (Absa, n.d.). Nedbank describes its cardless withdrawal service as allowing money to be sent to anyone with a South African cellphone number for withdrawal at a Nedbank ATM or participating store (Nedbank, n.d.). Standard Bank describes Instant Money as a voucher-based service involving a voucher number and PIN (Standard Bank, n.d.). This further shows why cardless withdrawls are popular due to their ease of use and low barriers to entry. For example you require almost no verification to withdraw funds, which makes it very fast and convenient, but can lead to major security and fraud related issues.

3.3 Abuse of Cardless Withdrawal in Fraud

The abuse of cardless withdrawal becomes clear when the service is placed inside a social engineering attack chain. In the scenario considered here, the scammer first creates fear and urgency. They may claim that the customer's bank account has been compromised, that a SIM swap has occurred, that a suspicious debit order needs to be reversed or that money must be moved to a safe place. The victim is then instructed to perform a transaction in their own banking app to ensure their money is safe, allowing the attackers to exploit and defraud them.

Standard Bank's public scam warning is highly relevant to this research proposal where it warns that fraudsters may call customers pretending to be from the bank, advise that there is fraud on the account and request that the customer send an Instant Money voucher to themselves. Once the customer receives the voucher, the customer is asked to read back the voucher details and PIN to validate the transaction (Standard Bank, n.d.). Standard Bank also describes a secure account and Instant Money vishing scam where victims are instructed to perform Instant Money transactions for authentication, debit-order reversal or unauthorised-transaction reversal (Standard Bank, n.d.). This directly supports the attack pattern addressed by this proposal.

The Prudential Authority's 2025 banking sector risk assessment report explains that instant money cash-out can allow users to send money to a South African cellphone number so the recipient can withdraw funds without needing a bank account (Prudential Authority, 2025). The report also lists immediate ATM or cash-send withdrawals as a banking red flag in certain financial-crime contexts (Prudential Authority, 2025). This lends support the idea that rapid cash-out is rising method of financial theft and fraud.

3.4 Fraud Detection and Anomaly Detection

Financial fraud detection is commonly treated as an anomaly or pattern recognition problem because suspicious behaviour is often hidden inside large volumes of normal or benign activity. Bolton and Hand (2002) describe this problem across areas such as card fraud, money laundering and telecommunications fraud. The same basic issue exists in cardless cash withdrawal fraud. A single cash-send transaction may not look suspicious on its own, but the surrounding behaviour

may change the interpretation of it. For example, a new recipient, an unusually high amount, a short time between login and voucher creation, or very fast voucher redemption may indicate that the transaction should be treated as higher risk.

The main difficulty is that fraud detection cannot rely on one simple rule or one clean dataset. Ngai et al. (2011) show that financial fraud detection uses many different data mining techniques because fraud occurs in different forms and through different channels. West and Bhattacharya (2016) also show why automated detection is needed, as manual review becomes impractical when transaction volumes are high. This is especially true in mobile banking, where suspicious activity may need to be detected before the transaction is completed or before a cardless withdrawal voucher is redeemed.

Another issue is that fraud data is rarely balanced or complete. Fraud cases are normally much fewer than legitimate transactions, fraud patterns change over time and labelled fraud data may not be available due to privacy around financial data (Abdallah, Maarof and Zainal, 2016).

Anomaly detection provides one way of dealing with this limitation. Chandola, Banerjee and Kumar (2009) define anomaly detection as the process of finding patterns that do not conform to expected behaviour. In this study, that means comparing a cardless withdrawal transaction against the customer's usual behaviour. Isolation Forest, proposed by Liu, Ting and Zhou (2008), is one possible model because it attempts to isolate unusual observations rather than requiring large numbers of labelled fraud examples. This could help identify transactions that are unusual in terms of amount, timing, recipient behaviour or redemption speed.

However, an anomaly score should not be treated as proof of fraud. A legitimate customer may send a relatively large amount to a new recipient for a valid reason, while a scam transaction may sometimes look fairly normal. For this reason, the proposed prototype should use a hybrid risk scoring approach rather than relying on one model alone. Rule-based systems can capture known scam indicators, such as a new recipient, rapid voucher redemption or a transaction close to the product limit. Anomaly detection can then add a customer-specific view of whether the transaction is unusual compared to previous behaviour. The final output should therefore be a risk score with alert reasons, not a hard claim that fraud has definitely occurred.

4 Methodology

4.1 Epistemological Stance

This research study will use a pragmatic epistemological stance. This is because the research and results of it are concerned with a real banking security problem and the development of a prototype to address this problem. The primary goal is to investigate whether behavioural risk scoring can provide useful real-time alerts for a specific fraud scenario. Additionally a synthetic dataset will be designed, generated and used to train and test the models that will be used to

detect the cases of fraud or benign behaviour. Therefore the results of this study will be dependant on the outputs and real world applications of said prototype

The research will follow a quantitative research approach. The results of this research will be dependant on the outputs, accuracy, precision and recall of the prototype that will be developed. These quantitative outputs will therefore decide the overall viability and feasibility of such an approach to combat banking fraud, with a special focus on cardless cash withdrawls.

The deliverable artefact will be a real-time behavioural risk scoring prototype for suspicious cardless cash withdrawal transactions. The prototype will not be a full bank fraud system. It will be a controlled system that simulates events, extracts features, calculates scores and triggers alert decisions.

4.2 Data Used

The study will use synthetic banking event data. This is necessary because real customer banking and fraud data is highly sensitive and unlikely to be publicly available. The use of synthetic data is also safer from a POPIA, GDPR or other privacy laws as well as from an ethics perspective because no real customers, accounts or transaction histories are processed.

The dataset will be designed to represent realistic mobile banking event sequences rather than only isolated transactions. This is important because the risk is often visible in the sequence of actions. For example, a suspicious journey may involve a login, immediate navigation to send-cash, creation of a new recipient, high-value voucher creation and rapid redemption.

Some of the features that may be considered are:

- customer_id - Customer identifier
- login_time - Time at which the mobile banking session starts
- transaction_time - Time at which the cardless withdrawal is created
- amount - Cardless withdrawal amount
- avg_customer_amount - Customer's historical average cardless withdrawal amount
- amount_ratio - Amount divided by customer average
- recipient_known - Whether the recipient number was used before
- previous_cardless_count - Number of previous cardless withdrawal transactions
- time_since_login - Minutes between login and transaction
- recent_profile_change - Whether profile, device or limit changed recently
- voucher_redeemed - Whether the voucher was redeemed
- redemption_delay - Time between voucher creation and redemption
- redemption_distance - Distance from usual redemption or transaction area

- risk_label - Normal, suspicious or fraud-like scenario label

4.3 Detection Approach

The prototype will investigate three detection approaches. The first approach will be a rule-based heuristic model. This will assign scores or values for known risk indicators. For example, a new recipient, a transaction near or over the daily limit, rapid redemption and a recently added device or profile change may add risk. This approach is explainable and closely matches how a simple fraud analysis rule might be described.

The second approach will be an anomaly detection model. Isolation Forest is the initial preferred model, although other models such as Local Outlier Factor or One-Class SVM are other considerations. The anomaly model will attempt to identify transactions that differ from normal customer behaviour. The model output will be an anomaly score which will then be used to guide a decision on whether the transaction was fraudulent or not.

The third approach will be a hybrid risk score. This will combine the rule-based score and anomaly score into one final risk value. This is the primary method that will be considered and evaluated because it allows the prototype to capture both known fraud indicators and unusual customer behaviour.

An example scoring structure may be:

- 0-39 Low Risk
- 40-69 Medium Risk
- 70-89 High Risk
- 90+ Critical Risk

The final thresholds may be adjusted during evaluation based on false positives, negatives, model accuracy and overall detection performance.

4.4 Final Prototype Deliverable

The prototype will be developed as a lightweight system that can process simulated event data and display the risk score of the transaction and alert outputs. Python will be used for feature extraction, modelling and evaluation. Scikit-learn will be used to train and evaluate the anomaly detection models. Specialised LLMs will be used to define, describe and generate the synthetic dataset required for such a research topic. Due to the nature of fraudulent, it is typical that the number of benign transactions will greatly outnumber fraudulent ones. It is for this reason the dataset will be be imbalanced towards benign cases. An interactive dashboard to highlight, explain and demonstrate the capabilities of the detection system will be the primary GUI to use and interact with the system.

The evaluation will compare the rule-based model, anomaly model and hybrid model. The main metrics will be precision, recall, false positive rate and detection rate. Accuracy will not be treated as the most important metric because

fraud-like behaviour will be rare in the dataset as previously discussed This follows the same general reasoning used in fraud and security detection problems: a model can look accurate by simply predicting the majority class, while still missing the cases that matter. The evaluation will also include scenario testing. Normal scenarios will include low-value cardless withdrawals to known recipients, occasional emergency withdrawals and quick but legitimate redemptions. Suspicious scenarios will include new recipient numbers, unusual amounts, rapid voucher redemption, multiple vouchers in one session and transactions occurring shortly after login or profile changes.

5 Planning

- June - Draft and submit research proposal
- July - Design, describe and generate synthetic dataset used for training and evaulating anomaly models
- August - Define heuristic rules and train anomaly models
- September - Gather and evaluate results of all three approaches described and tested
- September/October - Draft and submit final research paper

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Generative AI declaration form for assignment submission

Q1. Student Information

Name & Surname	Yeshlen Moodley
Student Number	17281904
Course Code	COS 700
Assignment Title	COS 700 Research Proposal

Q2. AI Declaration [tick one box]

1. I **used** Generative Artificial Intelligence in the current assignment	X
2. I did **not** use any Gen AI for the current assignment	

Ensure you are permitted to use generative AI by reading the instructions for your assignment or assessment, or by consulting your study guide. Failure to follow the instructions regarding using generative AI for your assignment constitutes academic dishonesty. Examples of generative AI include ChatGPT, Google's Gemini, Microsoft's Copilot, Claude, and Meta's Llama 3 AI integrated into WhatsApp and Facebook. If you selected option 1 in Q1, please continue to complete the rest of the form:

Q3. Prompts used

Please paste all the prompts you created for the assignment and indicate which aspect(s) of the assignment the specific prompt was used for.

AI Tool	Prompt	Aspect of Assignment
ChatGPT	What are the most effective ML models for anomaly detection	ML model training
ChatGPT	How can synthetic data be used for research and model training	Dataset creation
ChatGPT	What are common current fraud detection mechanism	Literature Review
ChatGPT	What banking services in South Africa have cardless withdrawal systems	Literature Review
ChatGPT	What is the pragmatic epistemological stance in simple terms	Research Methods

Q4. Type of usage

Briefly describe the features for which you used the Gen AI. You may say “Not Applicable” where needed:

Feature used	Description
Brainstorming and idea generation	I used this to help me decide on useful ML models
Language editing suggestions	N/A
Feedback and revision suggestions	
Explaining complex concepts	I used this to help understand different epistemological stances and how they worked. I also used it to explain what models are useful or anomaly detection
Writing coach	N/A
Other (please specify):	

Q5. Ethical use

Write a brief paragraph explaining how you ensured the usage of Gen AI was aligned with the ethical and responsibility requirements of the University of Pretoria (link). Consider examples such as repurposing and reintegrating ideas generated by Gen AI with your own thoughts, integrating Gen AI ideas with other literature, critically evaluating Gen AI outputs, maintaining transparency about Gen AI usage, enhancing your learning and ensuring comprehension despite using Gen AI, and personal development through using Gen AI as an assistant. If you directly used text or data generated by Gen AI, ensure it was cited appropriately.

Ethical use statement:

In using Gen AI I aimed to use it to aid me in completing my assignment. This involved explaining concepts or assistance in understanding useful ML models. I attempted to understand why it suggested something and used other sources and documentation to supplement my understanding.

Q6. Why did you use Gen AI?

I used Gen AI to aid me in understanding concepts I did not really understand as it can explain it to me real time and I could bounce off my understanding and ideas against it until I fully understood a concept.

Signature:

Student Signature	Y.Moodley
Date	13/06/2026

Note: The Department of Education Innovation would like to thank Prof. Celeste Combrinck and Ms. Nelé Loubser for their innovative ideas, which have been instrumental in creating our new form.